AUTOMATIC CLASSIFICATION OF CRIME-RELATED TEXT REPORTS USING NLP

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CHAPTER I: INTRODUCTION

1.1 Background Of The Study

A crime is an illegal act that is penalized by the state or another authority. The term "crime" refers to an offence that harms not only a single person but also a community, society, or state (Khan et al., 2022).

Criminal activity is present daily, in a multiplicity of illegal actions in several domains, such as drug trafficking, computer crime, and theft, to just mention a few examples. Upon the occurrence of a crime, criminal police investigators are in charge and start a set of actions to enable the construction of the so-called 'chain of custody' to identify the alleged culprits and to present them in court.(Carnaz et al., 2021).

Classification of Crime Reports is an important process in the analysis and management of crime-related data. In this Context: An ID, such as the type of crime, date and time of the incident, location, witnesses, and description of the incident(Hariguna & Ruangkanjanases, 2023)

Text classification specifically focuses on categorising and organising text data to facilitate easier management and analysis. These techniques leverage natural language processing, machine learning, and data mining to extract meaningful patterns and categorize text data. thereby transforming unstructured text into structured, actionable insights (Taha et al., 2024)

Crime-related data is exploding due to the constant reporting and distribution of crimerelated information on the internet in the rapidly evolving information technology and communication era. Online newspapers and social networks offer real-time information about what is happening in a specific location, at a particular moment, and with numerous cause(Ali et al., 2023).

Natural language processing is an area of computer science that is primarily concerned with transforming natural language into a formal language, such as a programming language, so that a computer can process and understand it, mainly combining algorithms. (Martínez Hernández et al., 2026).

Automated Crime Report Analysis is a technological approach used to automate the process of evaluating, sorting and understanding crime reports that enter the police system or relevant government agencies(Hariguna & Ruangkanjanases, 2023). The rise in cybercrime and the complexity of multilingual and code-mixed complaints present significant challenges for law enforcement and cybersecurity agencies. These organizations need automated, scalable methods to identify crime types, enabling efficient processing and prioritization of large complaint volumes. Manual triaging is inefficient, and traditional machine learning methods fail to capture the semantic and contextual nuances of textual cybercrime complaints(Rani et al., 2024).

The proposed research paper explores the application of machine learning techniques in crime analysis problem, specifically focusing on the classification of crime-related textual data. Through a comparative analysis of various machine learning models, including traditional approaches and deep learning architectures, the study evaluates the data (Mussiraliyeva & Baispay, 2024)

The 2021 amendment to South Korea's Criminal Procedure Law has significantly enhanced the role of the police as investigative authorities. Consequently, there is a heightened demand for advanced investigative expertise among the police, driven by an increase in the number of cases each investigator handles and the extended time required for report preparation. This situation underscores the necessity for an artificial-intelligence-supported system to augment the efficiency of investigators. (Park et al., 2024).

The ability to be anonymous on social media and the lack of adequate resources to fight cybercrime are catalysts for the rise in criminal activities, especially in South Africa. We proposed a system that will analyse and detect crime in social media posts or messages. The new system can detect attacks and drug-related crime messages, hate speech, and offensive messages.(Lombo Lombo et al., 2022).

The integration of machine learning techniques in crime analysis offers unprecedented opportunities to harness the vast amount of online textual data for enhancing public safety and security. By overcoming inherent challenges and leveraging the capabilities of ML algorithms, researchers and law enforcement agencies can gain invaluable insights into criminal behaviour and trends, ultimately contributing to more effective crime prevention and intervention strategies (Mussiraliyeva & Baispay, 2024)

In today's world, security is the most prominent aspect that has been given higher priority. Despite the rapid growth and usage of digital devices, the measurement of crimes in underdeveloped countries is still challenging(Umair et al., 2020)

The majority of Somalia's current crime reporting and management systems are manual. Labor-intensive and prone to human error. By creating an automated NLP framework especially suited for the Somali language, this study seeks to address these issues. It is anticipated that this transition from manual sorting to AI-driven analysis will greatly speed up law enforcement responses and enhance data integrity in the Somali security industry.

1.2 Problem Statement

Crime posts sent using digital channels (such as social media, or online forms) thought to be processed and categorised instantly in a perfect law enforcement setting. This enables prompt prioritisation and routing to the appropriate department (such as cybercrime, homicide, or theft), guaranteeing a prompt response that can save lives and stop additional criminal behaviour.

The majority of crime posts classifications are now done by hand. Every day, thousands of text-based posts must be evaluated by human dispatchers or officers in order to determine the type of crime. Today's massive amount of digital data makes this manual approach labour-intensive, slow, and prone to major delays.

The use of manual classification results in a number of serious problems: Slow Reaction: When a post is waiting in a queue, human error – fatigue brought on by information overload – causes posts to be misclassified or high-priority threats to be missed. Resource inefficiency: Instead of actively investigating and ensuring community safety, skilled staff devote hours to administrative sorting.

This study suggests using natural language processing (NLP) to create an automated classification system. The system will be trained to examine the linguistic patterns of crime posts, classify them automatically with high accuracy, and identify urgent instances for prompt attention by utilising machine learning methods. The goal of this shift from manual to AI-driven sorting is to significantly increase the speed of police and court responses.

1.3 Research Objectives

1.3.1 General Objective

The main objective of this research is to develop and put into use an intelligent automated framework that uses machine learning and natural language processing to transform the way Security agencies classify crime posts. This framework will replace manual sorting with high-Speed semantic analysis that guarantees operational efficiency and data integrity.

1.3.2 Specific Objectives

- To reduce the reliance on manual methods of classifying crime posts.
- To collect and preprocess crime-related text data for analysis.
- To extract and analyse key features from crime-related text data for accurate classification.
- To develop an automated NLP system for classifying crime-related text posts.

1.4 Research Questions

1. How to reduce the reliance on manual methods of classifying crime posts?
2. How to collect and preprocess crime-related text data for analysis?
3. How to extract and analyse key features from crime-related text data for accurate Classification?
4. How to develop an automated NLP system for classifying crime-related text posts?

1.5 Significance of the Study

The goal of this project is to create an AI-powered natural language processing system that can automatically categorize text posts pertaining to crimes, making it noteworthy. Researchers and law enforcement organizations can save time and money on manual data analysis by automating the classification process. Decision-making procedures may be improved; the system can help create intelligent crime monitoring tools, and it can increase the accuracy of identifying various sorts of crimes. Additionally, by showing how Natural Language Processing (NLP) may be used practically in crime analysis, this study advances the academic subject of NLP.

1.6 Scope of the Study

The design, development, and assessment of a system based on Natural Language Processing (NLP) that automatically ascertains whether a given text report is related to a crime or not is the main objective of this study.

1.7 Organization of the Study

I. **Chapter II:** Literature Review The Literature Review chapter is the summary of the study conducted by the researcher on the existing literature and studies that are associated with the topic of the study. This study reviews the existing research conducted to determine the effectiveness of crime detection and text classification using Natural Language Processing (NLP) techniques. Additionally, this chapter aims to identify the various machine learning algorithms and approaches that have been implemented on social media data to classify and identify crimes.

II. **Chapter III**: Methodology The third chapter discusses the methodology used for the research in this study. It shows how the social media data is collected and preprocessed. That is, in the stage of data preprocessing, the texts in social media posts are cleaned, tokenized, and the useful features are extracted. This chapter also provides more light about the machine learning algorithms that are employed in this study to classify the social media posts as crime and non-crime. Hence, the researcher fully explained data preparation and selection of the machine learning algorithm used.

III. **Chapter IV** System Analysis, Design And Implementation. This chapter will discuss the analysis, design, and implementation of the proposed system, which is an NLP system to classify social media posts. It will discuss the system architecture, data flow, text preprocessing, feature extraction, and classification of social media text. It will also show the flow of social media text from the input to the output of the system. It will also discuss the coding structure of the system and the deployment of the system.

IV. **Chapter V:** Conclusion and Recommendations This chapter concludes and summarizes all the findings from the study. It also discussed the validity of the experimental results. The study discussed how the NLP model is used to classify crimes on social media and some possible ways of improving it. Recommendations were made on how to improve the performance of the system. These include increasing the number of samples for training and re-training the model for accuracy. The system could also be integrated into existing policing platforms for effective crime control

CHAPTER II:LITERATURE REVIEW

2.1 Introduction

Crimes are the most common social issues nowadays, affecting the economic growth, quality of life, and economy of any country. Crimes affect the reputation of a country on an international scale and affect the economy of the country by placing a financial burden on the government in hiring additional police forces. For the eradication of crimes, the government needs to adopt an optimized strategy and sustainable e-governance information systems. Algorithms that predict the occurrence of crimes based on time and location can help the government to deploy law enforcement in highly dangerous areas (Umair et al., 2020)

Criminal activity is present daily, in a multiplicity of illegal actions in several domains, such as drug trafficking, computer crime, and theft, just to mention a few examples. Upon the occurrence of a crime, criminal police investigators are in charge and start a set of actions to enable the construction of the so-called chain of custody to identify the alleged culprits and to present them in court. These actions are multidisciplinary and, depending on the crime, they may involve distinct tasks, such as digital and biological forensics analysis, interviews with witnesses, and interrogations with suspects and other individuals who may be potentially implicated (Carnaz et al., 2021)

The classification of crime reports is a crucial process in the analysis and management of crime-related data. In this context, a crime report is written or digital information that contains various details about the crime incident, such as the type of crime, date and time of the incident, location, witnesses, and description of the incident. The goal of crime report classification is to group these reports into appropriate groups based on their characteristics, which facilitates efficient analysis, monitoring, and action by the authorities. One of the main benefits of classifying crime reports is their ability to aid law enforcement and investigation. By classifying crime reports by crime type, authorities can more easily identify potential crime patterns(Hariguna & Ruangkanjanases, 2023)

Crime-related data is exploding due to the constant reporting and distribution of crime-related information on the internet in the rapidly evolving information technology and communication era.Online newspapers and social networks offer real-time information about what is happening in a specific location, at a particular moment, and with numerous causes.These facts depict real-world events, such as accidents, innovations, conferences, natural disasters, or crime events. These resources have provided a substantial amount of crime data, which is the domain of their study, presenting a need to extract precise and valuable information to promote and strengthen the fighting capacities against crime capabilities, thus improving public protection and reducing the likelihood of future crime prospects. Moreover, information possession and effective deployment are essential in fighting crime and are in high demand from various end-users.(Ali et al., 2023)

Criminal detection and prediction constitute an essential practice in crime analysis, seeking to provide effective techniques for supporting law enforcement. Researchers develop diverse strategies and solutions utilizing machine learning (ML) and deep learning (DL) algorithms to foresee or detect criminal activities that have demonstrated considerable potential in addressing crime detection issues (Al-harbi & Alghieth, 2025)

Natural language processing is an area of computer science that is primarily concerned with transforming natural language into a formal language, such as a programming language, so that a computer can process and understand it, mainly combining techniques from AI, linguistics, statistics, and computing for the development of algorithms. It is a fundamental part of NLP as it focuses on understanding the meaning of words, phrases, or sentences and how they are used in a specific context.(Martínez Hernández et al., 2026)

Extracting events from news stories as the aim of several Natural Language Processing (NLP) applications (e.g., question answering, news recommendation, news summarization) is not a trivial task, due to the complexity of natural language and the fact that news reporting is characterized by journalistic style and norms. Those aspects entail scattering an event description over several sentences within one document (or more documents), applying a mechanism of gradual specification of event-related information.(Bonisoli et al., 2023)

Automated Crime Report Analysis is a technological approach used to automate the process of evaluating, sorting, and understanding crime reports that enter the police system or relevant government agencies. The main objective of this method is to improve efficiency and effectiveness in crime data management, as well as enabling authorities to respond to crime events faster and more accurately. Automated analysis of crime reports involves using technologies such as artificial intelligence and natural language processing to extract key information from incoming crime reports. (Hariguna & Ruangkanjanases, 2023)

The rise in cybercrime and the complexity of multilingual and code-mixed complaints present significant challenges for law enforcement and cybersecurity agencies. These organizations need automated, scalable methods to identify crime types, enabling efficient processing and prioritization of large complaint volumes. Manual triaging is inefficient, and traditional machine learning methods fail to capture the semantic and contextual nuances of textual cybercrime complaints.(Rani et al., 2024.)

The proposed research paper explores the application of machine learning techniques in the crime analysis problem, specifically focusing on the classification of crime-related textual data. Through a comparative analysis of various machine learning models, including traditional approaches and deep learning architectures, the study evaluates their effectiveness in accurately detecting and categorizing crime-related text data. The performance of the models is assessed using rigorous evaluation metrics, such as the area under the receiver operating characteristic curve (AUC-ROC), to provide insights into their discriminative power and reliability.(Mussiraliyeva & Baispay, 2024)

The 2021 amendment to South Korea’s Criminal Procedure Law has significantly enhanced the role of the police as investigative authorities. Consequently, there is a heightened demand for advanced investigative expertise among the police, driven by an increase in the number of cases each investigator handles and the extended time required for report preparation.This situation underscores the necessity for an

artificialintelligence-supported system to augment the efficiency of investigators. In response, their study designs a hybrid model that fine-tunes two Transformer-based pretrained language models to extract 18 key pieces of information from legal documents automatically.(Park et al., 2024)

The integration of machine learning techniques in crime analysis offers unprecedented opportunities to harness the vast amount of online textual data for enhancing public safety and security. By overcoming inherent challenges and leveraging the capabilities of ML algorithms, researchers and law enforcement agencies can gain invaluable insights into criminal behaviors and trends, ultimately contributing to more effective crime prevention and intervention strategies(Mussiraliyeva & Baispay, 2024)

In the contemporary era, ensuring security has become a critical priority due to the rapid proliferation of digital technologies and devices. As digital adoption increases globally, especially in developing regions, monitoring and addressing crime accurately becomes increasingly complex and challenging. Research suggests that developing countries face persistent cybersecurity challenges and limitations in effectively detecting and combating digital crime and security risks in these environments (Yadav et al., 2020)

In today's world, security is the most prominent aspect that has been given higher priority. Despite the rapid growth and usage of digital devices, the measurement of crimes in underdeveloped countries is still challenging(Umair et al., 2020)

2.2 Impact of crimes on social media

Social media representations of crime have the power to influence how the general public views crime, law enforcement, and the legal system. Inaccurate coverage of crime-related news can lead to mistakes, portrayal of crime as a necessity and a demand of the time, which makes people sympathetic to the criminal and, in a way, encourages crime. It also undermines public confidence in the government and gives power to those who have taken the law into their own hands. Hate crime is the overrepresentation of crimes committed by a certain segment of society against that segment, which has become a social media trademark and fosters hatred and negative perceptions toward that segment(Abhishek Kumar, 2025)

The Social media exposure of crime refers to the way crime-related content is shared, discussed, and highlighted on social media platforms. Social media is now a major tool for people to know what is happening around them. It is now a necessary part of everyone's life, even for students. As the use of social media as a platform for gathering information continues, individuals will encounter posts that are related to crime incidents. People will have their own insights based on the topic, which might affect them in some ways. Social media networks make it simple to spread both positive and negative emotions among individuals. Criminology students often recognize that social media can serve as an educational tool. By exposing users to real-time information about crime trends and safety issues, social media can enhance public awareness. Research indicates that platforms like Twitter and Facebook facilitate the rapid dissemination of crime-related information, potentially empowering communities to take preventative measures. Exposure to crime on social media can distort public perception, leading to fear and anxiety(Ezekwe, 2025)

According to the data, a large number of students share excessive amounts of information via social media and emails, which could endanger them. The majority of organizations have put in place multiple security measures to prevent information from leaking on social media. Cyberbullying has become widespread, particularly among those who are vulnerable to it. Social media is frequently the target of security threats such as hacking, spoofing, spamming, cyberstalking, Trojan horses, worms, viruses, and exposure scams; for this reason, government and ICT policies have addressed these problems. Higher education institutions must devote more funds to student education, awareness, research and development, IT security, and social media crimes(Benard et al., 2021).

Social media is substantially affecting the world's dynamics and proves to be particularly true when it comes to the legal realm.3 As a matter of fact, in this digital era, these massive online platforms give rise to new challenges and opportunities in relation to the existing framework of international criminal law (hereinafter 'ICL')(Ali, 2021)

The penetration of social media into nearly every aspect of contemporary life has reshaped patterns of communication, information consumption, and interpersonal interaction.

Beyond its role as a tool for social connectivity, digital platforms have emerged as influential spaces where narratives of crime circulate rapidly and where offenders increasingly exploit online affordances to facilitate unlawful behavior. Their study investigates the extent to which these platforms affect both criminal conduct and public interpretations of crime trends within modern societies. The purpose of the research is to analyze how social media environments enable planning, performance, and dissemination of criminal acts, while also shaping public emotions, risk perceptions, and trust in institutions responsible for maintaining public order(Rashid et al., 2025)

These days, people utilize a variety of social media platforms to share information with friends, family, and other people for official, business, and personal reasons. Cyberstalking is another major problem brought on by social media use. Cyberstalking has been recognized as an increasingly prevalent anti-social issue that impacts victims, educational institutions, and human society as a whole. Cyberstalking on social media must be detected by an intelligent system(Gautam & Bansal, 2022)

Numerous risks that have been recognized and categorized are present on social media, according to the International Journal of Network Security & Its Applications. Al Hasib categorized social networking threats into three categories: identity-related threats, which include phishing, friending malicious actors, profile squatting, stalking, and corporate espionage; privacy-related threats, which include the sharing of private information on social media; and information security threats, which include well-known security threats on social media like worms, viruses, and cross-site scripting. Similar to this, Fire et al. divided the threats associated with social media

(Umeugo, 2023).

In today's digital age, the widespread growth of the digital media sector and information technology has significantly improved communication flexibility through platforms such as Facebook, Twitter, Instagram, and others. These platforms, driven by Web 2.0 technologies, enable global connectivity, fostering social interaction and community building beyond geographical constraints. Despite their advantages, such as accessibility and interactivity, social media also pose risks like cybercrimes and threats to security, making it crucial to evaluate their impact on societal security in the United Arab Emirates (UAE). The research problem focuses on how social media simultaneously affects societal security, aiming to identify strategies to mitigate risks while preserving its benefits (Alnaqbi & Ali, 2025)

2.3 Traditional Classification techniques in crimes

Traditional methods of crime prevention and detection have played an essential role in maintaining law and order within communities. These methods are characterized by their reliance on human intervention, community participation, and well-established policing practices. This section discusses these key traditional methods, highlighting their strengths and limitations(Apene et al., 2024)

The efficacy of several machine learning models, such as LSTM, GRU, Random Forest, SVM, and Logistic Regression is examined in their study. Based on thorough testing, the Random Forest, SVM, and Logistic Regression models performed well on tasks involving the classification of crimes. Traditional inspection of a large number of crime reports, however, may be laborious and prone to human error. However, the conventional manual classification of crimes based on textual descriptions is laborious, prone to human error, and frequently inconsistent(Eldho,2023)

Moreover, the manual annotation of some visual content can be problematic from an occupational health and safety perspective, as the potential exists for individuals to be exposed to a large number of indecent and disturbing images. To further compound matters, forensic practitioners generally have limited background knowledge of big data, Artificial Intelligence, and data sciences to help create models tailored for their own practices. In order to make full use of large forensic image databases and remove associated practical issues, specialized high-performance and accurate computer vision (e.g., automatic recognition and content filtering) and data analytics tools are required(Taha, 2024)

Clustering is also another common technique used to predict patterns of geographic criminal history and behavior. Remond and Baveja proposed a case-based reasoning (CBR) system that filtered out police cases for better prediction. It worked on noisy data and explained how police reports show only individual criminal activities and do not address gang crimes. Social networks are also used as a potential source of criminal activity indicators. Sadhana and Sangareddy used Twitter information to find real-time criminal offences. They also used the same data set to find the concentration of crime occurrences, and find large scale of hotspots. Thus, several techniques are used for analyzing crimes(Hossain et al., 2020)

From high-level categories to lower-level subtypes, the conventional classification of crime types is based on a hierarchical framework. This tree-based classification lacks tricategory semantic linkages since it recognizes different types of crimes as mutually independent when they do not branch from the same higher-level category. In any event, the prevailing trend is on the initial classification of crime types offered by each nation or local organization, which consists of hierarchical structures from a single point of view, ranging from high-level categories to lower-level subtypes. When crime types are not mutually independent, this conventional tree-based classification recognizes them as such. branch from the same higher-level category; hence, there are no semantic connections across categories. Even though they may be closely linked, crime types that originate from distinct high-level categories are handled very differently. For example, the crime type "disturbing the peace" and the crime type "liquor—drinking in public" may be perceived as inherently related since, from a police standpoint, the two offenses are related. However, according to the city of Boston's crime classification, "disturbing the peace" falls under the category of "disorderly conduct," while "drinking in public" falls under the category of "liquor violation."(Crivellari & Ristea, 2021)

Police in the United Kingdom use a hierarchical framework of UK Home Office codes, classes, sub-classes, and offence categories to categorize crime events. These classifications correspond to the Notifiable Offence List, which, for each class, corresponds to a pertinent law that is used to prosecute people who are thought to have committed a specific crime. For example, code 28/3, class "Burglary," subclass "Residential Burglary," offence category "28F Attempted Burglary— Residential," and its application related to Section 9 of the Theft Act (1968) are used to record an attempted residential burglary. The aforementioned categories were created to facilitate a variety of criminal justice procedures and allow for comparisons in crime statistics both within and between jurisdictions. However, even if this method of categorization is essential for the criminal justice system's administrative requirements, crimes may not adequately describe the ecological aspects of real-world criminal issues. Simply put, not all offenses of a given classification are created equal. Although classifying them as such makes administrative sense, it might not be the best way to comprehend the kinds of criminal opportunities and behaviors that arise in a given context to cause crime problems (Birks et al., 2020)

Using text mining, natural language processing, and machine learning approaches, our system tackles the problem of classifying unstructured legal documents by organizing and analyzing this data and extracting reports related to crime scenes. Our work stands out because we created a crime lexicon with 151 related terms and 70 crime instruments that were taken from different forensic sources. Despite being regarded as rather modest, it helped provide positive categorization results. To enhance crime scene investigation, the suggested crime dictionary can be expanded, used in sophisticated searches, and linked with police databases. (Bifari et al., 2024)

For a limited portion of the crime data sets, K-Nearest Neighbor (KNN), Support Vector Machines (SVM), decision trees, random forests, Naive Bayes, and so forth are comparatively more effective. The fundamental framework of crime prediction is shown in Fig. 1. A plan based on conventional machine learning is presented in Part (a). First, a training set and a test set are created from the data set. In order to achieve an appropriate representation of the crime data, the algorithm's fundamental requirement is to extract the crucial information that can represent the crime event data. Since there are many various kinds of crime data, it is vital to choose pertinent attribute data and then extract all of the data's features at once to increase the data's capacity for effective expression. The crime is then forecasted using a range of machine learning algorithms, including support vector machines, decision trees, and linear regression models.(Yin, 2022)

2.4 Cybercrime detection in social media

Cybercrime, to put it simply, is any criminal offense that uses a computer, digital devices, or the internet as a conduit. It falls under the category of contemporary crimes brought about by developments in the Internet and related technology. They might be categorized as hybrid offenses because, in contrast to more conventional crimes like robbery and theft, which can be clearly localized in terms of time and place of occurrence, these crimes may have a connection that is challenging to pinpoint. Given the aforementioned, some authors have characterized it as a composite crime that includes cyber-dependent crimes made possible by the Internet and digital technology, as well as cyber-enabled crimes, including organized crime, fraud, forgery, theft, and money laundering(Chinedu et al., 2021)

Social media is being used for a variety of activities, but it can also lead to cybercrime. Online sharing of personal information can lead to crime. Victims may be reluctant to report crimes due to triviality, embarrassment, or lack of awareness. Social media monitoring can be used to augment conventional crime reporting. Social media is being used to facilitate criminal activity, much like other emerging technologies and communication channels. It is imperative to protect private data being transferred via networks (Abdalrdha et al., 2023)

As with all cybersecurity and cybercrime issues, awareness is critical to help combat cybercrime on social media. Employees and organizations risk falling victim to cybercrime on social media. Awareness is also required to avert inadvertent perpetration and participation in cybercrime on social media. Organizations must incorporate cybercrime on social media awareness into general cybersecurity and cybercrime awareness training. Organizations must also measure their employees' cybercrime on social media awareness levels to gauge the effectiveness of their awareness training programs. Few studies exist on cybercrime awareness on social media, and most cybercrime awareness measurements do not consider cybercrime on social media(Umeugo, 2023)

According to the National White Collar Crime Center's "Criminal Use of Social Media" white paper, social media's recent rise has significantly altered the communication environment. Social media sites like Instagram, Facebook, LinkedIn, YouTube, and Twitter are used daily by millions of individuals. People interact with each other via various sites. These channels are used for advertising, hiring new employees, and routine communication within the public sector(Kaur et al., 2024)

Online social networks (OSNs) such as Twitter are being used more and more in today's society to spread information globally. Online social networks have gained popularity because of their broad reach, streamlined dialogues, anonymity, and capacity to deliver brief bursts of information. additionally could go into a channel for different forms of online harassment and bullying. Common conversations and banter on social networking sites can occasionally turn explosive, leading to the unchecked expression of hate, harassment, and aggressiveness without the possibility of personal danger or reciprocal physical harm. Reports of mental and psychological health problems in recent years have been directly connected to cyberbullying, abuse, and harassment. Quickly identifying posts meant for abuse and expunging them from the network to lessen user exposure is necessary to stop these abusive activities(Osho et al., 2020)

The proliferation of digital communication platforms has led to the emergence of cyberbullying as a major social issue. In contrast to conventional bullying, cyberbullying enables offenders to reach victims at any time and from any place, transcending physical boundaries. Particularly common among teenagers and young adults, this type of harassment has been connected to a number of detrimental effects, such as anxiety, sadness, and even suicidal ideation(Altayeva et al., 2024)

The findings show that many students share too much information through emails and social media, which may pose a threat to them. Most organizations have implemented multilayered security mechanisms information security may leak on social sites.

Cyberbullying has spread widely, especially among those who are prone to the practice of cyberbullying. Security attacks such as hacking, spoofing, spamming, cyber stalking, Trojans, worms, viruses, and exposure scams are common through social media, hence the ICT policy and government policy have addressed such issues. Higher learning institutions need to invest more resources in training students, awareness, research and development, IT security, and social media crimes. (Benard et al., 2021)

Threats from cybercrime are becoming more frequent every day. Due to a lack of record keeping at relevant offices for a variety of reasons, including victims' presumptions about police response, users' ignorance of IT (information technology) acts on cybercrimes, and victims' incapacity to identify that they have been victimized, there is currently no foolproof, systematic, and trustworthy tool for reviewing cybercrimes. Reducing and categorizing cybercrimes may also be aided by promoting and educating the public about them, as well as by identifying and maintaining area-specific cybercrime rates (Ch et al., 2020)

2.5 Challenges in the Classification of Crime-Related Text

Developing highly effective classifiers for such imbalanced datasets remains a significant challenge. As the dataset imbalance increases, the model tends to overfit the majority class, impairing its ability to achieve the desired performance on minority classes and leading to biased outcomes. Hence, the reliability of the model in real-world applications deteriorates, resulting in failures in critical scenarios where the accurate detection of minority classes is essential. (Taskiran et al., 2025)

Multi-label text classification involves assigning more than one class to a text based on analyzing the terms in the text body. presents an open-world multi-label text classification. In this type of task, the possible classes are not available a priori. The method needs to perform topic modeling for subsequent text classification. So, it fragments the texts and sends them to an LLM for keyword extraction. The words are grouped, and the centroids of each group form the labels for text classification. The labels for the text classification procedure can all be available a priori. However, some researchers investigate the need for a dynamic update of the possible labels available for classification(Vieira et al., 2025)

Text classification can be done manually, but it is time-consuming and has a high cost. Manual classification comes with lots of errors and is less accurate because of human error and a lack of domain knowledge. There was a huge revolution in text classification when machine learning models, such as the Support Vector Machines (SVM), Naive Bayes (NB), and Random Forest (RF) started to replace manual work, because these models not only reduce the time and cost but also are highly accurate for classification. Since the inception of machine learning models, numerous studies have been conducted to enhance, optimize, and refine the text classification process (Palanivinayagam et al., 2023)

The entire legal text is then divided into several predefined logical segments using structural characteristics. Next, the corresponding entities are extracted from each logical segment using a hybrid method of rules and deep learning. For simple entities, the rules and patterns are generated in the process of knowledge modeling to extract entities. For complex entities, the pre-trained model Bidirectional Encoder Representations from Transformers (BERT) is introduced into the task of legal text information extraction, and is combined with Bi-directional Long Short-Term Memory (Bi-LSTM) and Conditional Random Field (CRF) algorithms to extract entities. It is effective in solving the problem of polysemy in legal texts (Ren et al., 2022)

Beyond the analyses of free-text recorded by police, several studies have also explored how LDA topic models might be used to analyze crime-related phenomena discussed in unstructured content posted online. Combine LDA and collaborative representation classifiers in an attempt to detect ‘criminal intention’ in free-text extracted from online blogs. Relatedly, applies LDA topic models to geotagged tweets posted to the online platform Twitter in Chicago to generate neighborhood dominant topics indicative of the ecology of a particular location. These classifications are then used to positively augment a kernel density estimation-based crime prediction algorithm that seeks to estimate the levels of distinct offences in the immediate future (Birks et al., 2020)

A large language model (LLM) is a class of artificial intelligence models, typically based on deep learning, specifically designed to comprehend and generate human language. These models are characterized by their enormous size, often containing hundreds of millions or even billions of parameters, which allows them to perform a wide range of natural language understanding and text generation tasks. For crime prediction tasks, an ML model can be trained on inputs to output labels from a fixed set of classes. However, recent advances in LLMs such as BERT and GPT-3 have introduced a novel approach known as prompting. In prompting, there is typically no need for further training; instead, the model's input contains a task-specific text called a prompt (Sarzaeim et al., n.d.)

Securing large language models (LLMs) and NLP technology in general has not been a priority until recently. Yet mass adoption of this technology has led to deployment in contexts where security failures present a risk to individuals, organizations, and society at large—demonstrated by, inter alia, LLM-assisted identity fraud phishing campaigns and automated influence operations If the latest NLP technologies are to withstand an increasing barrage of threats, NLP practitioners must now educate themselves on cybersecurity and cybersecurity practitioners must educate themselves on NLP (Lent et al., 2025)

As another example, if there is a spike in traffic accidents during a festival period, local police could investigate the causes and allocate appropriate healthcare resources to the affected area. If drunken driving is a significant cause, police could dispatch corresponding real-world statistics provided by administrative authorities. It conducted a case study on the efficacy of the news classification models using two reputable online news sources and seven prevalent types of crimes/accidents in Thailand(Tuarob et al., 2025)

Additionally, the issue of creating software tools and methods to aid in the investigation and prevention of criminal acts based on textual material disseminated online is still unsolved today. Numerous applications are available to handle the problem, such as hate voice recognition, crime prediction, criminal pattern modeling, identification of crime-related topics, etc. However, more effective techniques for analyzing criminal content in computer-mediated communication (CMC) are still required in order to prevent and solve crimes. Crime-related event extraction (CREE) is one of the most difficult and effective methods for identifying and forecasting criminal activity. The possibilities to gather organized data on a crime that has occurred or is likely to occur, compile this data, and then model a crime pattern are responsible for the success of the CREE job. In the meantime, it is still very difficult to extract different kinds of crime-related events (CRE) from texts that are not police or witness narrative reports, even with the presence of studies pertaining to the CREE task. Additionally, materials produced in low-resource and under-annotated languages make the Event Extraction process more challenging(Khairova et al., 2023)

2.6 Related Work

In recent years, there has been a lot of interest in using artificial intelligence (AI) and machine learning (ML) to analyze crime (Hariguna & Ruangkanjanases, 2023). created a model for an adaptive decision-support system that is specifically for automatically analyzing and sorting crime reports in the context of e-government. Their study built an automated framework that used Natural Language Processing (NLP) techniques to pull out important information from reports that came in. The system was found to greatly improve the efficiency of managing crime data, allowing authorities to respond to crime events more quickly and accurately. Looking at things together (Mussiraliyeva & Baispay, 2024).tested different machine learning methods - old styles and newer deep learning setups - to sort through text about crimes. Rather than making assumptions, they turned to reliable methods - such as the Area Under the Receiver Operating Characteristic Curve (AUC-ROC) - to assess which model performed best.

This shift led to tangible insights for police departments, revealing patterns in offender behavior along with hints about future actions. These findings didn't just show new tools - they helped shape smarter ways to stop crimes before they grow. Looking closely at tough investigation issues. (Park et al., 2024). created a mixed system to help South Korean law enforcement deal with new pressures after changes to the criminal procedure law in 2021 made cases harder to solve. Instead of relying on manual review, their approach used two existing transformer-based models that had already been trained on legal data - tweaked slightly through additional learning - to pull out 18 crucial details from paperwork automatically. Looking back, it became clear how vital AI tools are when investigators face growing workloads and longer report prep intervals.When it comes to sorting public crime data, (Ali et al., 2023). suggested using a BERT-driven system to boost how news documents about crimes are sorted. Because clear information and swift deployment matter in crime prevention, their work highlighted the need for precise data pulling by different user groups. It turned out that using existing language models led to much stronger narrative handling - especially

for stories tied to criminal activity - than older approaches did. Focusing first on materials not in English, (Carnaz et al., 2021).faced little data by building a labeled set of crime-linked Portuguese texts. Their effort showed how complex crime exams truly are - blending fields like digital enforcement, interview handling, and keeping evidence trails intact. This dataset came together to build a base resource - key for creating NLP and machine learning tools that handle intricate crime cases across languages beyond English. Looking into better training data, Liu et al. (2023) explored ways to boost text variety while keeping results accurate. Instead of just adding more samples, they tried swapping words with synonyms, tossing in random terms, or reshuffling full sentences. These shifts helped spread out the examples used to teach models. Because of this mix, the system learned to recognize patterns across deeper layers of meaning in crime classification. Over time, their findings pointed to stronger performance when faced with complex structures within criminal taxonomy. Starting with the challenge of uneven data spread, Franklin et al. (2024)

introduced a new approach - "text-guided mix-up" - aimed at handling the long-tail issue during sorting jobs. Drawing on how classes are linked in meaning, thanks to insights from pre-trained vision-language tools such as CLI P, they tapped into written guidance to ease imbalances shaped by sparse example sets. Looking at things this way fits well when studying crime patterns - some reported incidents happen much less often but still matter just as much in understanding what's happening. Looking at how African languages work, Sani et al. (2025), along with Alabi's group two years earlier, studied what happens when language models adjust through a method called LAFT - using a model named AfriBERTa, among others. Looking into Sani and team's work, they explored how people feel in Hausa using text analysis. Their findings show that tweaking big language models trained on multiple languages works better when taught only on African dialects. Since building a Somali crime news tool matters, it proves machines like AfriBERTa - built to adapt - can handle smaller languages well instead of just copying from English-based systems. Though rare, such shifts matter most where data runs thin.

Author & Year	Description	Methods	Limitations
Hariguna & Ruangkanjanases (2023)	Developed an adaptive decision support system for e-government crime analysis.	Automated NLP framework to extract key info from incoming reports.	Traditional ML methods fail to capture semantic and contextual nuances.
Mussiraliyeva & Baispay (2024)	Evaluated effectiveness in detecting and categorizing crimerelated text.	Comparative analysis of traditional vs. deep learning models.	Challenges in harnessing the vast amount of online textual data.
Park et al. (2024)	AI system to help South Korean police handle increased workloads.	Hybrid model finetuning two Transformer-based pretrained models.	Extended time required for manual report preparation remains a burden.
Ali et al. (2023)	Aimed to strengthen fighting capacities against crime through data extraction.	BERT-driven system to boost news document sorting.	Exploding volume of crime-related data makes manual possession difficult.

Carnaz et al. (2021)	Focused on non-English (Portuguese) crimelinked text classification.	Construction of a labeled dataset for forensic tasks.	Scarcity of labeled data in languages other than English.
Liu et al. (2023)	Explored data diversity in hierarchical multilabel classification.	Methods: Synonym replacement, random insertion/deletion/swap.	Excessive data enhancement may lead to model over-fitting.
Franklin et al. (2024)	Addressed long-tailed label distribution in training.	Text-guided mix-up technique using semantic relations.	Traditional deep learning requires costly, large, balanced datasets.
Sani et al. (2025)	Investigated sentiment analysis	Language-Adaptive Fine-Tuning (LAFT) using AfriBERTa.	Smaller languages often just copy English-based

2.7 Gap and Contribution

The reviewed literature provides strong evidence for automated crime-text analysis, but it offers limited support for Somali-language and code-mixed content and rarely connects classification to an operational investigation workflow. This study addresses that gap by: (1) preparing and evaluating a Somali-oriented labelled dataset; (2) preventing data leakage through a train-only TF-IDF fit; (3) comparing multiple supervised classifiers with emphasis on F1-score and crime-class recall; (4) deploying the selected model through a reproducible service architecture; and (5) integrating classification with monitoring, alerts, blacklist checks, reporting, and case management. The contribution is an applied decision-support system, not an autonomous policing or guilt-determination tool.

2.8 Proposed System

The proposed system is an NLP (Natural Language Processing) based text classification system that classifies any given social media post as crime-related or non-crime-related. The dataset used for this proposed system consists of social media posts, which are manually labelled by humans as crime-related or non-crime-related. The output is generated after the preprocessing of the input text and then passing it to the machine learning and deep learning algorithms. The algorithms used in this proposed system are Support Vector Machine (SVM), Random Forest, and BERT. The preprocessed features of the input text are used by these algorithms to classify and find out whether the input text is crime-related or not. The proposed system is trained and tested using a manually labelled dataset, and the performance is evaluated by calculating the accuracy, precision, recall, and F1-score. The proposed system has a simple web interface developed in Python and Flask. The interface allows users to input any social media post that needs to be classified. The proposed system then classifies and provides the output of whether the input social media post is crime-related or not, along with the confidence. The proposed system thus facilitates an efficient monitoring of the social media posts as compared to the manual surveillance required in traditional systems.

CHAPTER III: METHODOLOGY

3.1 Introduction

In Somalia and across the Horn of Africa, law enforcement agencies receive an increasing volume of unstructured crime-related text through social media platforms, messaging applications, citizen tip lines, and online news outlets. Unlike structured database records, these reports arrive as free-form Somali and English text that must be read, interpreted, and prioritized by human analysts before any investigative action can begin. The manual nature of this workflow creates bottlenecks that delay response to urgent threats, increase the risk of misclassification due to fatigue, and consume skilled personnel who could otherwise focus on field investigation.

The primary objective of this research is to design and implement an intelligent automated framework—BAREAI (Bare AI Crime Intelligence Platform)—that classifies text reports as crime-related or not crime-related using Natural Language Processing (NLP) and supervised machine learning. The system extends beyond pure classification by integrating Somali-language hybrid enrichment, Facebook page monitoring, blacklist matching, investigation case management, and operational reporting within a single web-based platform suitable for security agencies operating in resource-constrained environments.

This chapter presents the complete methodology adopted in developing BAREAI. It describes the system at a functional level, explains the three-tier architecture connecting the React frontend, Node.js backend, Flask AI microservice, and MongoDB database, and documents each major feature in detail. The chapter further outlines the machine learning workflow—from dataset collection and preprocessing through model selection, training, evaluation, and deployment—and concludes with the development environment and hardware/software requirements necessary to reproduce the system.

The methodology follows a structured software engineering lifecycle combined with a standard CRISP-DM (Cross-Industry Standard Process for Data Mining) approach for the machine learning components. Data understanding and preparation precede model experimentation; the best-performing classifier is serialized and integrated into a production inference service. Parallel to model development, agile iterations were used for full-stack implementation, allowing frontend, backend, and AI services to evolve concurrently while maintaining clear API contracts between layers. By documenting architecture, features, algorithms, and environmental requirements in this chapter, subsequent chapters can focus on system analysis, detailed design, implementation evidence, and empirical results without repeating foundational methodological decisions. This structured presentation aligns with graduation thesis conventions at Jamhuriya University of Science and Technology (JUST) and mirrors best practices observed in comparable early-warning systems developed for predictive analytics in East African contexts.

3.2 System Description

BAREAI is an NLP-driven crime text classification and intelligence platform developed to reduce reliance on manual sorting of digital crime reports in Somalia. The system accepts user-submitted content through multiple input modalities—direct text entry, URL submission, document upload (PDF, DOCX, TXT), and batch file processing—and returns a binary classification decision (Crime or Not-crime) accompanied by a confidence score, extracted keywords, detected locations, and a human-readable explanation of the decision rationale.

At the core of the platform lies a supervised machine learning classifier trained on a manually labelled dataset of crime-related and non-crime-related text samples. The training pipeline, implemented in the Jupyter Notebook Automatic_crime.ipynb, compares Support Vector Machine

(SVM), Random Forest (RF), and Bidirectional Encoder Representations from Transformers (BERT) architectures. Following comparative evaluation on accuracy, precision, recall, F1-score, and Area Under the ROC Curve (AUC-ROC), the best-performing model is serialized as crime_model.pkl together with its TF-IDF vectorizer (vectorizer.pkl) and served through a lightweight Flask REST API on port 5001.

The Node.js Express backend (port 5000) orchestrates the end-to-end analysis pipeline. Upon receiving a classification request, it validates input, extracts text from URLs or uploaded files when necessary, forwards normalized text to the AI service, applies hybrid Somali keyword and location enrichment rules, checks entries against a configurable blacklist of Facebook pages, persons, keywords, and websites, persists results to MongoDB, and triggers email notifications and in-app alerts when crime is detected or blacklist matches occur. This enrichment layer addresses limitations of purely statistical models when processing low-resource languages and domain-specific criminal vocabulary. Beyond reactive analysis, BAREAI provides proactive monitoring through an automated Facebook page scanner built with Puppeteer. Administrators configure a watchlist of suspicious pages; the backend periodically scrapes recent posts, classifies content using the AI model with keyword-based fallback when the AI service is unavailable, and surfaces detections through the Notifications module. Detected crime-related content can be escalated into formal investigation cases assigned to investigators with status tracking from open through under investigation to closed with verdict.

The React 19 frontend, built with Vite, delivers role-based interfaces for administrators, investigators, and analysts. Features include an analytics dashboard with KPIs and trend visualizations, the Analysis Center for multi-modal input, blacklist management, notification inbox, case management workflow, and PDF report generation. Authentication uses JSON Web Tokens (JWT) with email-based one-time password (OTP) verification, ensuring secure access to sensitive law enforcement data.

In summary, BAREAI transforms fragmented manual crime text triage into an integrated, semi-automated intelligence pipeline. It combines the speed and consistency of machine learning with domain-specific Somali linguistic rules, operational workflows, and social media monitoring capabilities that existing generic NLP tools do not provide.

3.3 System Architecture

The BAREAI system architecture follows a modular three-tier design that separates presentation, application logic, and data persistence while isolating the machine learning inference engine as an independent microservice. This separation of concerns improves maintainability, allows independent scaling of the AI component, and enables the backend to continue operating with keyword-based fallback if the AI service is temporarily unavailable.

The presentation layer consists of a single-page application (SPA) built with React 19 and Vite, served on port 5173 during development. The SPA communicates exclusively with the backend REST API over HTTPS, attaching JWT bearer tokens to authenticated requests. Client-side routing manages navigation between public pages (landing, login, registration) and protected pages (dashboard, analysis, blacklist, notifications, cases, reports, profile) based on user role.

The application layer is implemented in Node.js using Express 5. It exposes RESTful endpoints grouped by domain: authentication (register, login, OTP verification, password reset), analysis (text, URL, file, batch), blacklist CRUD operations, investigation cases, notifications, reports, dashboard statistics, and Facebook monitoring control. Middleware handles JWT verification, role-based access control (RBAC), request validation, error handling, and CORS configuration. Mongoose ODM provides the object-document mapping layer for MongoDB.

The AI service layer runs as a Python Flask application on port 5001. At startup, it loads the serialized scikit-learn model and TF-IDF vectorizer from disk using joblib. The primary endpoint POST /predict accepts a JSON body with a text field, vectorizes the input, runs inference, and returns the predicted label, confidence score, and probability distribution. This stateless design allows horizontal scaling behind a load balancer in production deployments.

The data layer uses MongoDB database crime_detection_system with collections for users, analysis history, blacklist entries, investigation cases, notifications, and Facebook monitoring configuration. Document-oriented storage suits the semi-structured nature of analysis results, which include variable-length keyword arrays, nested location objects, and metadata fields that would require complex relational schemas.

External integrations include Facebook page scraping via Puppeteer (headless Chromium), email delivery through Nodemailer for OTP and alert notifications, and HTTP fetching of remote URLs via Axios and Cheerio for content extraction. Figure 3.1 illustrates the complete architecture and inter-layer communication paths.

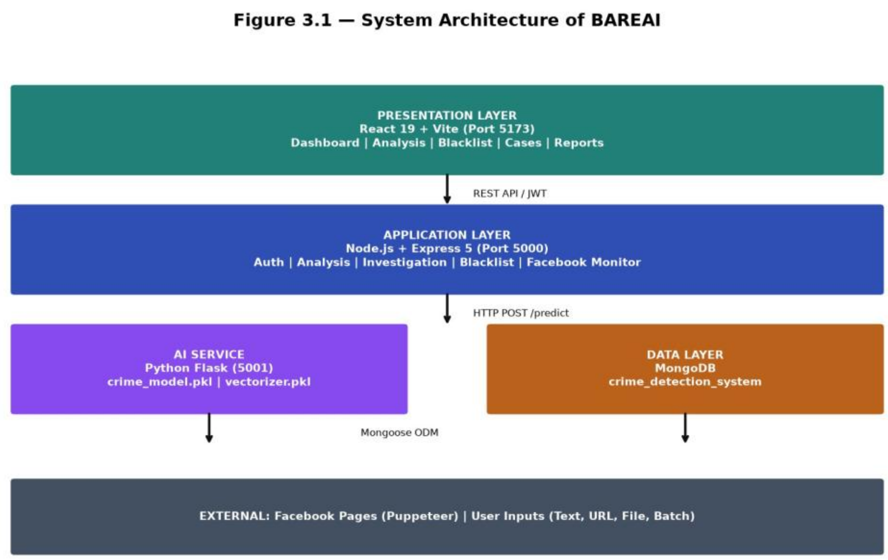

Figure 3.1 - System Architecture of BAREAI

3.4 System Features

The BAREAI platform incorporates a comprehensive set of features designed to support the full lifecycle of crime text intelligence—from initial detection through investigation and reporting. Each feature was designed with input from the problem domain (Somali law enforcement workflows) and implemented to be accessible to users without deep technical expertise. The following subsections describe each feature in detail, including its purpose, user interactions, and underlying technical implementation.

3.4.1 Authentication and Authorization

User authentication in BAREAI employs a secure email-and-password registration flow with email OTP verification to confirm account ownership before activation. Upon successful login, the backend issues a signed JWT containing the user identifier, email, and role (admin, investigator, or analyst). Tokens expire after a configurable period; expired tokens return HTTP 401, prompting re-authentication.

Registration with email verification via one-time password

Secure login with bcrypt-hashed password storage

JWT-based session management for stateless API access

Role-based access control: admin (full access), investigator (cases + analysis), analyst (analysis+reports)

Password reset flow with email OTP

Profile management and account settings

3.4.2 Crime Text Analysis

The Analysis Center is the primary operational interface for crime text classification. Users select an input modality—Text, URL, File, or Batch—and submit content for analysis. The backend normalizes input, invokes the AI model, applies hybrid enrichment, checks blacklist matches, saves results to history, and returns a structured response displayed in the UI with color-coded Crime/Not-crime badges, confidence percentage, keyword chips, location tags, and explanatory text.

Direct text input with real-time character count and Somali/English support

URL analysis: fetch remote page, extract visible text, classify extracted content

File upload: PDF (pdf-parse), DOCX (mammoth), TXT plain text extraction

Batch mode: upload CSV/TXT with multiple entries for sequential classification

Hybrid enrichment: 9 Somali regex keyword patterns + location extraction

Blacklist override: matched watchlist entries force isCrime = true

3.4.3 Analytics Dashboard

The admin dashboard provides at-a-glance operational intelligence through key performance indicators (KPIs) and interactive charts. Metrics include total analyses performed, crime detections, safe classifications, detection rate percentage, recent alerts, top detected keywords, and time-series trends. The dashboard enables supervisors to assess system utilization and crime detection patterns without querying raw database records. KPI cards: total analyses, crimes detected, safe reports, detection rate Line/bar charts for crime vs. safe trends over time .Top keywords word cloud or ranked list from recent detections .Recent alerts feed with quick navigation to full analysis details Role-restricted: primarily admin and analyst access

3.4.4 Blacklist Monitoring

The blacklist module maintains a watchlist of entities associated with criminal activity or requiring heightened scrutiny. Entry types include Facebook pages (name + URL), persons (name + optional description), keywords (terms that trigger alerts), and websites (URLs). When analysis content matches a blacklist entry—through exact or partial string matching—the system flags the result as crime-related regardless of ML prediction and generates an immediate alert.

CRUD operations for blacklist entries with type categorization

Automatic matching during every analysis request

Integration with Facebook monitoring for page-specific surveillance

Email and in-app notification on blacklist match

Admin-only management with audit trail in analysis history

3.4.5 Investigation Case Management

When crime is detected, authorized users can escalate an analysis result into a formal investigation case. Cases include title, description, linked analysis reference, assigned investigator, priority level, status (open, under investigation, closed), and final verdict. Investigators receive assignment notifications and can update case status, add notes, and record verdicts. This workflow provides human-in-the-loop validation essential for law enforcement accountability.

Case creation from analysis results or manual entry

Officer/investigator assignment with notification

Status workflow: Open → Under Investigation → Closed

Verdict recording: confirmed crime, false positive, inconclusive

Case listing with filters by status, priority, and assignee

3.4.6 Notifications and Alerts

The Notifications module centralizes all system-generated alerts in a unified inbox. Alert types include crime detections from user analysis, Facebook monitoring hits, blacklist matches, and case assignments. Each notification contains a timestamp, severity indicator, summary text, and deep link to the related analysis or case. Users can mark notifications as read and filter by type or date.

Real-time crime detection alerts with confidence and source

Facebook monitoring scan results with post excerpts

Case assignment notifications for investigators

Read/unread status tracking

Pagination and search for high-volume alert environments

3.4.7 Reports and Data Export

Operational reporting enables export of analysis history and statistics as PDF documents. Report types include general summary reports, individual analysis detail reports, weekly activity reports, and monthly trend reports. Reports incorporate charts, tables, and formatted text suitable for supervisory review or archival. The reporting module uses server-side PDF generation to ensure consistent formatting across clients.

General report: overall statistics and recent activity

Individual report: single analysis with full metadata

Weekly and monthly aggregated trend reports

PDF download with institutional formatting

Filter by date range and classification outcome

3.4.8 Facebook Page Monitoring

Proactive social media monitoring complements reactive user-submitted analysis. Administrators add Facebook page URLs to the monitoring configuration. A background service using Puppeteer launches headless Chromium, navigates to each configured page, extracts recent post text, and classifies content using the AI model. When the AI service is unreachable, a fallback classifier uses 40+ Somali and English crime-related keyword patterns. Detected posts create notifications and optional analysis history entries.

Configurable page list with scan interval (default 60 seconds)

Puppeteer-based scraping without Facebook Graph API dependency

AI classification with keyword fallback for resilience

Duplicate detection to avoid repeated alerts for same post

Start/stop monitoring controls for administrators

3.5 System Methodology

This section details the machine learning and software engineering methodology followed in developing BAREAI. The workflow encompasses dataset acquisition, preprocessing, feature extraction, model selection and training, evaluation, deployment, and integration with the full-stack application. Each step was documented in the Jupyter Notebook Automatic_crime.ipynb and validated against standard NLP classification practices.

3.5.1 Dataset

The training dataset (dataset.csv.csv) comprises manually labelled text samples representing crime-related and non-crime-related content relevant to the Somali security context. Labels were assigned by domain experts who reviewed each text sample and classified it as crime-related (label = 1) or not crime-related (label = 0) based on presence of criminal activity descriptions, threats, violence references, illegal transactions, or other indicators defined in the labelling guidelines.

The dataset includes a mix of Somali and English text reflecting real-world report language patterns, including code-switching between languages within single documents. Source types represented in training data mirror production inputs: social media posts, news excerpts, citizen tip simulations, and formal report narratives. The raw dataset is stored in CSV format with at minimum two columns: text (the raw input string) and label (binary target variable).

Class distribution was examined during exploratory data analysis. Stratified splitting preserves the observed label proportions across training, validation, and test partitions, while supported classifiers use balanced class weights. No synthetic samples are introduced into the held-out test data. Data quality assurance involved manual review of a random sample of labelled entries to verify inter-annotator consistency. Ambiguous samples—such as news reports about crime that do not constitute direct criminal reports—were excluded or re-labelled according to predefined guidelines distinguishing informational content from actionable crime intelligence.

Table 3.*1 - Dataset Schema*

Attribute *Attribute*	**Description**	**Data Type**	**Example**
text	Raw report or post content	String	Nin hubaysan ayaa toogtay...
label	Binary classification target	Integer (0/1)	1 = crime-related
language	Primary language (optional)	Categorical	Somali / English / Mixed
source	Origin category (optional)	Categorical	social_media / news / tip

3.5.2 Data Preparation

Data preparation transforms raw text into numerical feature representations suitable for machine learning algorithms. The pipeline implemented in Automatic_crime.ipynb follows established NLP preprocessing stages adapted for Somali-English mixed content.

3.5.2.1 Text Cleaning

Remove HTML tags, URLs, email addresses, and excessive whitespace. Normalize Unicode characters and standardize quotation marks. Preserve Somali diacritics and characters essential for keyword matching.

3.5.2.2 Lowercasing

Convert all text to lowercase to reduce vocabulary dimensionality. Applied consistently at both training and inference time to prevent train-serve skew.

3.5.2.3 Tokenization

Split text into word tokens using whitespace and punctuation boundaries. Somali morphological complexity is handled implicitly through n-gram features rather than explicit stemming, as Somali stemmers are less mature than English equivalents.

3.5.2.4 Stop Word Removal

Optional removal of high-frequency low-information words. For Somali text, a custom stop word list was used alongside English stop words from scikit-learn.

3.5.2.5 TF-IDF Vectorization

Term Frequency-Inverse Document Frequency (TF-IDF) converts tokenized text into sparse numerical vectors. Parameters include max_features (vocabulary size limit), ngram_range=(1,2) for unigrams and bigrams, min_df (minimum document frequency), and max_df (maximum document frequency to exclude overly common terms).

3.5.2.6 Train-Validation-Test Split

The prepared dataset is divided using stratified sampling into 70% training, 15% validation, and 15% held-out test partitions. TF-IDF is fitted only on the training partition and then used to transform validation and test data. The validation set supports feature and model selection, while the untouched test set provides the final unbiased evaluation.

3.5.2.7 Class Balancing

Class distribution is preserved through stratified splitting. Models that support class weighting, including Logistic Regression, Linear SVM, Decision Tree, and Random Forest, are configured with balanced class weights so that minority-class errors receive appropriate attention. The held-out test distribution is not synthetically altered.

3.5.2.8 Outlier Detection

Extremely short texts (fewer than 5 characters) and duplicate entries are removed. Texts exceeding maximum length thresholds are truncated to prevent memory issues during BERT fine-tuning.

3.5.3 Model Selection

Model selection evaluates multiple supervised classification algorithms to identify the best performer for crime text classification. Three models were selected based on literature review findings (Chapter II), computational feasibility, and interpretability requirements for law enforcement deployment: Support Vector Machine (SVM), Random Forest (RF), and BERT.

3.5.3.1 Support Vector Machine (SVM)

Support Vector Machine is a discriminative classifier that finds the optimal hyperplane separating crime-related and non-crime-related text samples in high-dimensional TF-IDF feature space. SVM maximizes the margin between classes, providing good generalization on text classification tasks with sparse features (Joachims, 1998; Islam et al., 2020).

For BAREAI, SVM with a linear kernel was trained on TF-IDF vectors due to computational efficiency and strong baseline performance documented in cybercrime and hate speech detection literature. Hyperparameter tuning via GridSearchCV explored C (regularization strength) values including 0.1, 1, and 10. Cross-validation on the training set guided parameter selection before final evaluation on the held-out test set.

SVM advantages for this project include fast training and inference (critical for real-time analysis), low memory footprint for the serialized model, and deterministic predictions. Limitations include difficulty capturing long-range semantic dependencies that transformer models handle more effectively, and reduced performance on highly imbalanced datasets without balancing techniques.

3.5.3.2 Random Forest (RF)

Random Forest is an ensemble learning method constructing multiple decision trees on bootstrap samples of the training data and aggregating their predictions through majority voting. For text classification, RF operates on the same TF-IDF feature matrix as SVM but captures non-linear feature interactions through hierarchical splits (Breiman, 2001).

Hyperparameters tuned for BAREAI include n_estimators (number of trees: 100, 200), max_depth (maximum tree depth: None, 20, 50), and min_samples_split (minimum samples required to split a node: 2, 5). RF provides intrinsic feature importance scores, enabling analysis of which terms most strongly influence crime classification decisions—valuable for investigator trust and model explainability. RF demonstrated strong test accuracy in experiments, though training accuracy reaching 100% in some configurations suggested potential overfitting, addressed through depth limits and cross-validation. RF inference latency remains acceptable for the sub-30-second response time requirement.

3.5.3.3 Transformer Model Consideration (BERT)

BERT is a pre-trained transformer language model capturing bidirectional contextual word representations (Devlin et al., 2019). Fine-tuning BERT for crime text classification leverages transfer learning from large-scale language pre-training, potentially improving performance on Somali-English mixed text where TF-IDF bag-of-words representations lose contextual meaning. BERT was reviewed as a possible contextual benchmark, but it was not the deployed model in the completed implementation. The production pipeline uses TF-IDF and conventional supervised classifiers because they provided strong measured performance with substantially lower training and inference requirements. Future work may evaluate multilingual or African-language transformer models under the same leakage-free split and held-out test protocol.

BERT offers superior semantic understanding compared to traditional bag-of-words methods but requires greater computational resources for training and inference. Model size exceeds scikit-learn serialized models, influencing the deployment decision. If BERT test performance marginally exceeds SVM/RF without justifying latency and infrastructure costs, the simpler model is preferred per Occam's razor for operational deployment.

Table 3.*2 - Model Comparison Criteria*

Criterion *Criterion*	**SVM**	**Random Forest**	**BERT**
Feature type	TF-IDF sparse	TF-IDF sparse	Contextual embeddings
Training speed	Fast	Moderate	Slow (GPU preferred)
Inference speed	Very fast	Fast	Moderate
Interpretability	Moderate	High (feature importance)	Low
Somali support	Via n-grams	Via n-grams	Via multilingual pre-training

3.5.4 Model Training and Evaluation

The dataset was divided using stratified sampling into 70% training, 15% validation, and 15% held-out test partitions. TF-IDF was fitted only on the training text to prevent data leakage. Baseline models and feature configurations were compared on validation data; selected models were then tuned through three-fold stratified cross-validation on the combined training and validation partitions. Final performance was measured once on the untouched test set. Evaluation included accuracy, weighted precision, weighted recall, weighted F1-score, crime-class recall, confusion matrix, and ROC-AUC.

The tuned Random Forest achieved the strongest held-out performance in the final experiment: 92.3% accuracy, 92.3% weighted F1-score, and 95.9% recall for the crime-related class. Logistic

Regression and Linear SVM achieved weighted F1-scores of 91.5% and 91.3%, respectively. Random Forest was therefore selected for deployment, and the classifier, fitted TF-IDF vectorizer, and model metadata were serialised as crime_model.pkl, vectorizer.pkl, and model_meta.pkl.

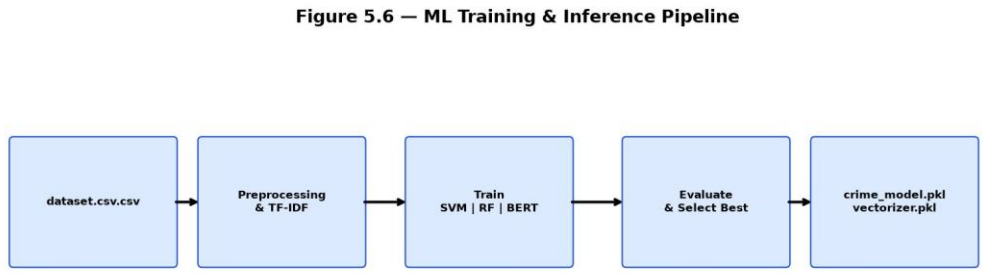

Figure 3.*2 - ML Training and Inference Pipeline*

3.5.5 Hybrid Enrichment and Inference

Production inference extends beyond raw ML predictions. The Flask AI service returns model output; the Express backend then applies a hybrid enrichment pipeline: (1) Somali keyword regex matching against 9 crime-related patterns, (2) location extraction from known Somali place names, (3) blacklist entity matching, and (4) confidence threshold adjustment. If any enrichment rule triggers, isCrime may be set to true even when the ML model predicts non-crime with moderate confidence, improving recall for domain-critical vocabulary not well-represented in training data.

3.5.6 System Implementation Methodology

Full-stack implementation followed an iterative agile approach with two-week sprints. Sprint 1-2 focused on AI model training and Flask API. Sprint 3-4 implemented Express backend with authentication and analysis endpoints. Sprint 5-6 developed React frontend pages. Sprint 7-8 added Facebook monitoring, blacklist, cases, and notifications. Sprint 9-10 conducted integration testing, bug fixes, and documentation.

API contracts between frontend, backend, and AI service were defined using JSON request/response schemas documented in Chapter IV. MongoDB schemas were designed using Mongoose with validation rules, indexes on frequently queried fields (userId, createdAt, isCrime), and references linking analyses to cases and notifications.

3.6 System Development Environment

Development was conducted on Windows 10/11 workstations with Visual Studio Code as the primary integrated development environment. Version control used Git with feature branch workflow. Local development runs three concurrent services: Flask AI (python ai-model/app.py), Express backend (npm run dev), and Vite frontend (npm run dev). MongoDB Community Edition runs locally or via MongoDB Atlas for cloud development.

Jupyter Notebook provided the interactive environment for exploratory data analysis, preprocessing experiments, and model training in Automatic_crime.ipynb. Python virtual environment (venv) isolated ML dependencies from system Python. Node.js npm managed JavaScript dependencies for both frontend and backend packages.

Table 3.3 *- Development Tools*

Tool *Tool*	**Version**	**Purpose**
Visual Studio Code	Latest	Code editor and debugging
Jupyter Notebook	7.x	ML experimentation
Git	2.x	Version control
Postman	Latest	API testing
MongoDB Compass	Latest	Database inspection
Chrome Dev Tools	Latest	Frontend debugging

3.7 System Requirements

This section describes the hardware and software required for the development, operation, and maintenance of the BAREAI (Bare AI Crime Intelligence Platform) system. These requirements are presented in two tables—one for hardware and one for software—to demonstrate that the system can operate efficiently in resource-constrained environments.

3.7.1 Hardware Requirements

specifies minimum and recommended hardware configurations. The deployed TF-IDF and Random Forest pipeline can operate on a multi-core CPU; additional memory improves concurrent analysis and monitoring. GPU acceleration is not required for the production model, although it would be useful for future transformer experiments.inlagusoodaro

Table 3.*4 - Hardware Requirements*

Component *Component*	**Minimum**	**Recommended**	**Purpose**
Processor	Intel Core i3 / AMD Ryzen 3	Intel Core i5+ / AMD Ryzen 5+	ML training, Puppeteer scraping
RAM	8 GB	16 GB or higher	Model loading, concurrent browser instances
Storage	128 GB HDD	512 GB SSD	Datasets, models, logs, MongoDB data
GPU	Not required	NVIDIA GTX 1060+ (6GB VRAM)	BERT fine-tuning acceleration
Network	Broadband 10 Mbps	Broadband 50+ Mbps	API calls, Facebook scraping, email
Display	1366×768	1920×1080	Dashboard visualizations

3.7.2 Software Requirements

This section explains the minimum software requirements for using our program, as shown in the table below.

Table 3.*5 - Software Requirements*

Software *Software*	**Version**	**Purpose**
Windows	10 or 11	Development and deployment OS
Python	3.9+	ML training and Flask AI service
Node.js	18 LTS+	Express backend runtime
MongoDB	6.0+	Document database
Google Chrome	Latest	Puppeteer headless browser
Microsoft Word	2019+	Thesis documentation

3.7.3 Required Libraries and Frameworks

Table 3.6 lists the primary libraries and frameworks used across the BAREAI stack. Python packages are installed via pip from requirements.txt in the ai-model directory. Node.js packages are managed via package.json in frontend and backend directories respectively.

Table 3.6 - Required Libraries and Frameworks

Library / Framework *Library / Framework*	**Layer**	**Purpose**
scikit-learn 1.8.0	AI	SVM, RF, TF-IDF, metrics, train-test split
joblib	AI	Model and vectorizer serialization
Flask	AI	REST API for inference
transformers (optional future evaluation)	AI	BERT fine-tuning (Hugging Face)
pandas / numpy	AI	Data manipulation
Express 5	Backend	HTTP server and routing
Mongoose	Backend	MongoDB ODM
jsonwebtoken / bcryptjs	Backend	Authentication
Puppeteer	Backend	Facebook page scraping
Axios / Cheerio	Backend	URL fetching and HTML parsing
pdf-parse / mammoth	Backend	Document text extraction
Nodemailer	Backend	Email notifications
React 19	Frontend	UI components
Vite	Frontend	Build tool and dev server
React Router	Frontend	Client-side routing
Recharts	Frontend	Dashboard charts

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